

Clinical Trials

A Hands-on Workshop

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What we will do today

1. Clinical trial Methods: A Recap
2. Sample size calculation
3. Random allocation
4. Clinical trial registration
5. Learning from mistakes
6. Discussion and ideas

Clinical trial Methods: A Recap

The essentials

- **Type of trial:** parallel, multi-arm, cross-over, etc
- **Unit of allocation:** individual, cluster
- **Framework:** superiority, non-inferiority
- **Allocation:** Non-random, Quasi-random, Random
- **Controlled:** not, placebo, active
- **Phase:** Only for drug trials

Sample size calculation

What a researcher should do

- Estimate the number of participants needed to achieve study objectives
- Describe how it was determined
- Indicate clinical and statistical assumptions supporting calculations.

The null hypothesis

$$H_0 : P_t - P_c = 0$$

Practical exercise

- Clinical question
 - Among ICU patients with septic shock and AKI without urgent need for dialysis, does an early, compared to delayed, initiation of renal replacement therapy reduce all-cause mortality at 90 days?

Arguments

- the expected proportion for the treatment group
- the expected proportion for the control group
- the equivalence limit
- the required study power
- the desired alpha level
- sided.test

Clinical and statistical assumptions

- `treat = .4`
- `control = .5`
- `delta = 0`
- `power = .8`
- `alpha = .05`
- `sided.test = 2`

Calculation

- Tools
 - use our online tool <https://ecebm.shinyapps.io/Power/>
 - use epiR package in R
 - `epi.sssupb(treat = .4, control = .5, delta = 0, n = NA, power = .8, alpha = .05, sided.test = 2)`
- The estimated sample size is 770
 - Treatment arm: 385
 - Control arm: 385

Random allocation

Simple random allocation

- Equal probability
- Unpredictability
- No guarantee of balance

Random allocation rule

- Forced equal allocation
- Vulnerability to selection bias late in trial

Block randomization

- Balanced allocation within blocks
- Protection against temporal trends
- Predictability risk with fixed block sizes

Tools

- use R functions
- use our online tool <https://ecebm.shinyapps.io/randalloc>

Clinical trial registration

Open access primary registries

- Example
 - EU Clinical trials register
 - UK's Clinical study registry
 - PACTR
 - Clinicaltrials.gov
- Mandatory: minimum dataset for trial registration
- Use SPIRIT for the protocol
 - Perfectly aligns with trial registries

Learning from mistakes

Comments: RCT

1. Selection bias
2. Serious contradiction in retinopathy inclusion criterion
3. Only one HbA1c measurement at the end of follow up
4. Intervention was not standardized
5. Composite outcome problem
6. Follow-up Duration

Discussion and ideas
